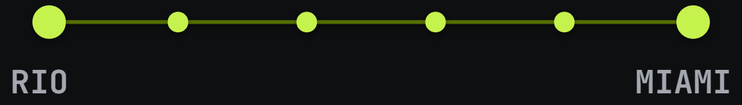
TESTNET · TEST KEYS · NOT REAL CUSTODIANS

Radox Custody Rail

Every handover of a shipment, signed by both hands, on Stellar.

Dual-signed custody for goods in transit · First corridor: Rio–Miami

Scan: the live shipment
rio-miami-rail.vercel.app



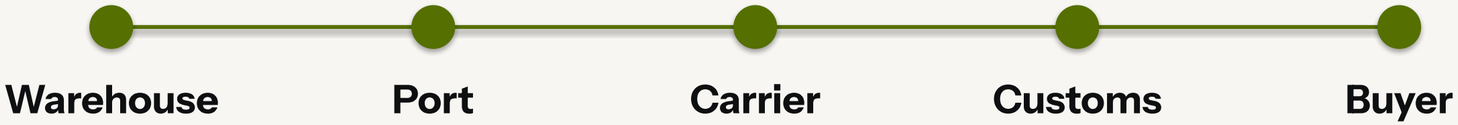
Hack Meridian / 2026 LISBON | GUAVA® Stellar

© 2026 HackMeridian · Lisbon

PMF · THE PROBLEM

Who holds the cargo right now?

Between seller and buyer, a shipment changes hands several times.



Each handover is recorded on paper or inside one operator's own system.

Our hypothesis: a financier cannot see who holds the goods before paying. No customer has confirmed it yet; we are testing it with mentors.

US$2.5T

Global trade finance gap.

ADB Global Trade Finance Gap Survey, Dec 2025. It sizes the financing gap; it is not evidence of the custody problem.

Context: our explainer video about a Rio–Miami trade-corridor initiative poses one possibility: a unique digital record for every shipment that both sides can check. This project is a custody layer for that record.

© 2026 HackMeridian · Lisbon · #HackMeridian26

THE SOLUTION

Every handover, signed by both hands

Outgoing custodian
`signs`

One Stellar transaction
`custody_registry.handover`

Incoming custodian
`signs`

Either signature missing → the call fails. Both sign, or nothing changes.

- Each handover moves custody and extends an on-chain hash chain, the `head`.
- The token stays with its holder in transit; only custody changes.
- After Delivered, the operator burns the token with the existing `redeem`[1].

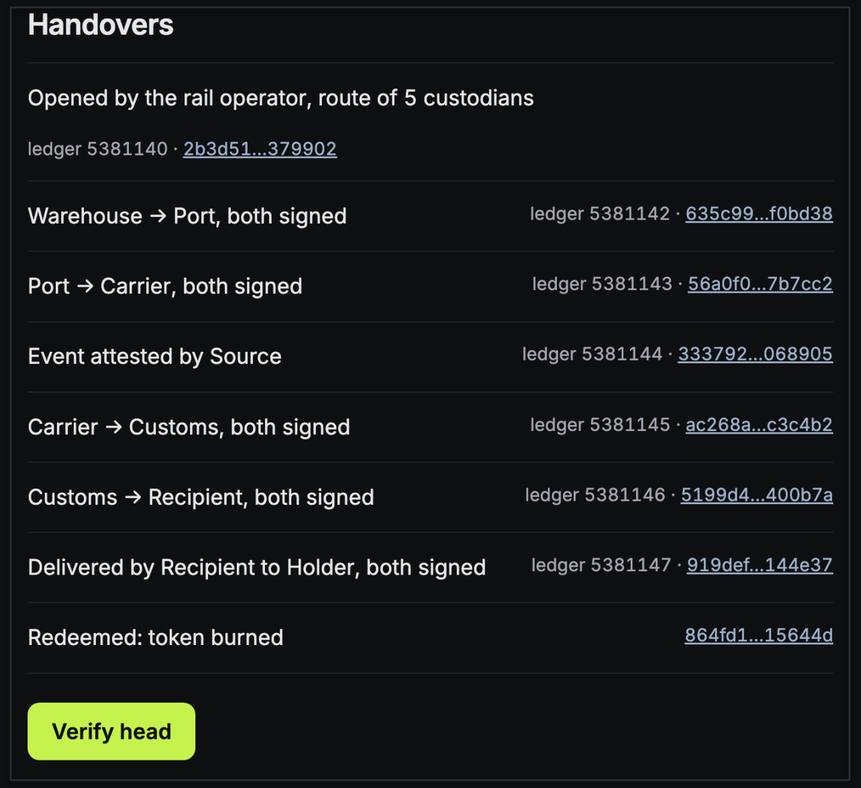
A custody record keyed to the container number. In production it would sit beside the carrier's bill of lading; not integrated in this build.

[1] Radox platform policy, not enforced on-chain by the registry; redeem still needs the holder's signature.

© 2026 HackMeridian · Lisbon · #HackMeridian26

THE PRODUCT · LIVE ON TESTNET

Anyone can check the chain. No login.



One public page per shipment: every handover, who signed it, and its Stellar transaction.

Verify head recomputes the hash chain in the browser and compares it with the contract.

MATCH

recomputed from 7 transactions, checked live 25 Oct 2026

Reads Stellar RPC and Horizon directly; read-only.

rio-miami-rail.vercel.app

© 2026 HackMeridian · Lisbon · #HackMeridian26

TECHNICAL WORKFLOW

7 transactions on Stellar testnet. All successful.

open	handover 1	handover 2	handover 3	handover 4	deliver	burn
operator	Warehouse → Port	Port → Carrier	Carrier → Customs	Customs → Recipient	Recipient → Holder	existing redeem
2b3d51f0	635c99ac	56a0f006	ac268a49	5199d4fb	919def8c	864fd164

7
successful testnet transactions

9
invalid calls refused (simulated)

24
contract unit tests pass

MATCH
verify-head rebuilds the chain

`custody_registry` CCBJVXMJYWPHTDJHDFPRGKPGXGEP24JOPJAMUS5C3BMZJTRTWR2H4LII

Each hash opens at stellar.expert/explorer/testnet/tx/<hash>; full hashes in the repo's proofs/lifecycle.json.

All six actors are test keys held by one harness: this proves the mechanism, not independent custodians.

© 2026 HackMeridian · Lisbon · #HackMeridian26

ECOSYSTEM · WHY STELLAR

Three Stellar features do the work

01	**Two signers, one call**	Soroban authorises both custodians inside one contract invocation: the handover is atomic.
02	**Events plus an on-chain head**	RPC keeps contract events about 7 days. The head stored in the contract lets anyone rebuild the full history later.
03	**The token stays put, then burns**	The shipment token never moves in transit; the existing Radox redeem burns it after delivery.

Nearest neighbours we found (LumenLoop, SCF, Scout; searched 24–25 Oct 2026): Kutana and Trustless Work release escrow on trade milestones. None records a physical handover signed by two custodians, and no SEP covers it.

Open source, Apache-2.0: a custody primitive other Stellar real-world-asset issuers can reuse.

© 2026 HackMeridian · Lisbon · #HackMeridian26

USERS · BUSINESS MODEL

Custodians sign. Financiers check.

CUSTODIANS	Warehouse, port, carrier, customs: sign each handover.
EXPORTER · BUYER	See where the goods are and who holds them.
FINANCIER	Checks the custody chain before paying.
OPERATOR	A financial institution, our client, runs the rail for its trade flows.

MODEL · THE OPERATOR PAYS

SWIFT-style: the record format is public; Radox runs the rail and the API.

Revenue: a fee from the financial institution that operates the rail. Not priced yet.

Radox is the rail, never the issuer. Regulation and licences stay with the client and its local counsel.

© 2026 HackMeridian · Lisbon · #HackMeridian26

HACKATHON PROGRESS

Built this weekend, on disclosed prior work

PRIOR WORK · BEFORE SUN 25 OCT 09:00

— Radox platform and 3 Soroban contracts on Stellar mainnet (public repo, created 2026-08-19)

— transfer_agent and its redeem (new testnet instance from the existing wasm)

— Planning files, 24–25 Oct

BUILT AT HACKMERIDIAN

— custody_registry contract, 24 unit tests

— Lifecycle harness: 7 testnet transactions, 9 refused calls

— verify-head script and the public shipment page

— Radox API reads the registry: same head (closed backend, testnet)

AI attribution Code, tests and scripts were written with AI coding agents (Claude Code); the team decided scope and what ships. Per-file record in AI_USAGE.md. The video narration is an AI voice generated from Pedro's voice.

© 2026 HackMeridian · Lisbon · #HackMeridian26

NEXT · THE ASK

Next: one real shipment, then mainnet

NOW

Testnet this weekend: contract, 7 transactions, public page, API read.

NEXT

Full testnet test matrix. A first pilot shipment with one producer (none agreed yet).

PLANNED

SCF Build application (not yet submitted). Security audit before mainnet (not done).

FUTURE

Mainnet. A USD on-ramp leg (not built).

No users or revenue yet. First users come from one pilot shipment in the Rio–Miami corridor.

Our ask **Introductions to one financial institution and one logistics operator for a first pilot shipment, and a Stellar engineer's review of the design.**

© 2026 HackMeridian · Lisbon · #HackMeridian26

THE TEAM

Why us

Pedro Wakigawa Saragossy

Founder and team captain. Built Radox: 3 Soroban contracts on Stellar mainnet.

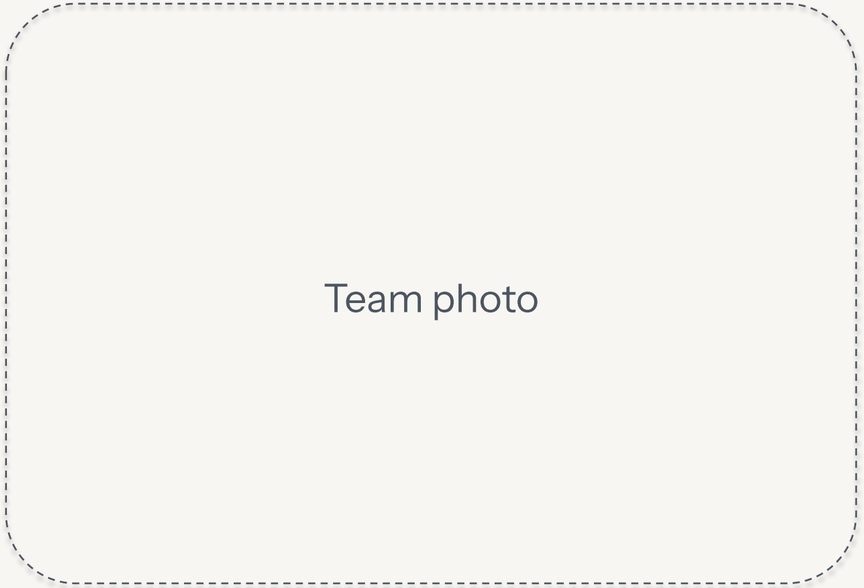
Leo Schlanger

Developer. Led custody_registry, the testnet harness and the proofs this weekend.

Vhendala

QA and product review: reads every deliverable as a judge would.

Leo and Vhendala joined for the hackathon.



Live page rio-miami-rail.vercel.app **Code** github.com/pwsaragossy/rio-miami-rail (Apache-2.0)

AI Code written with AI coding agents (Claude Code). Narration: an AI voice from Pedro's voice. No AI-generated images in this deck.

© 2026 HackMeridian · Lisbon · #HackMeridian26